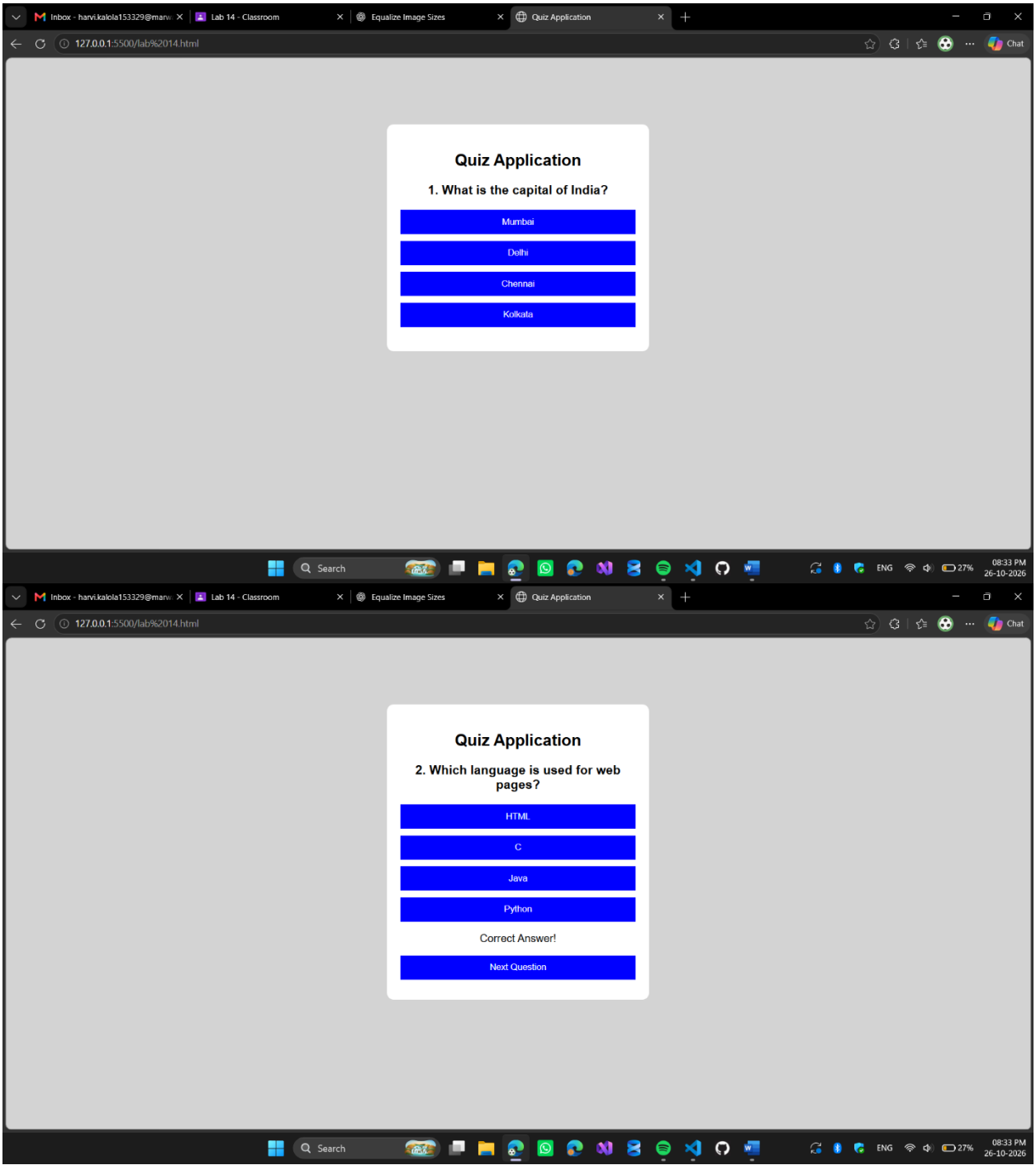
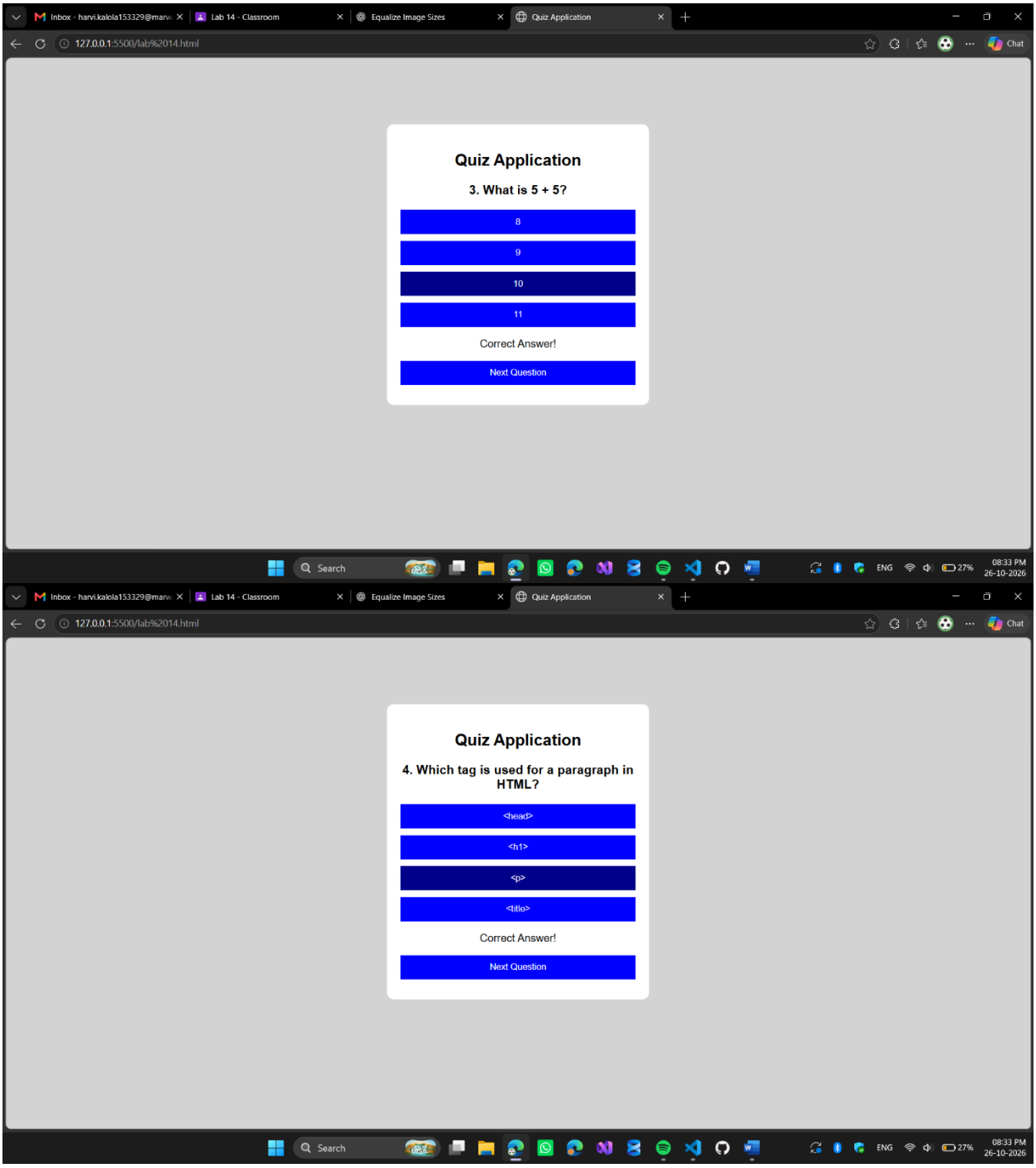
Lab#14 Create a quiz application that displays one question at a time. Track the user's score and show the final result at the end.

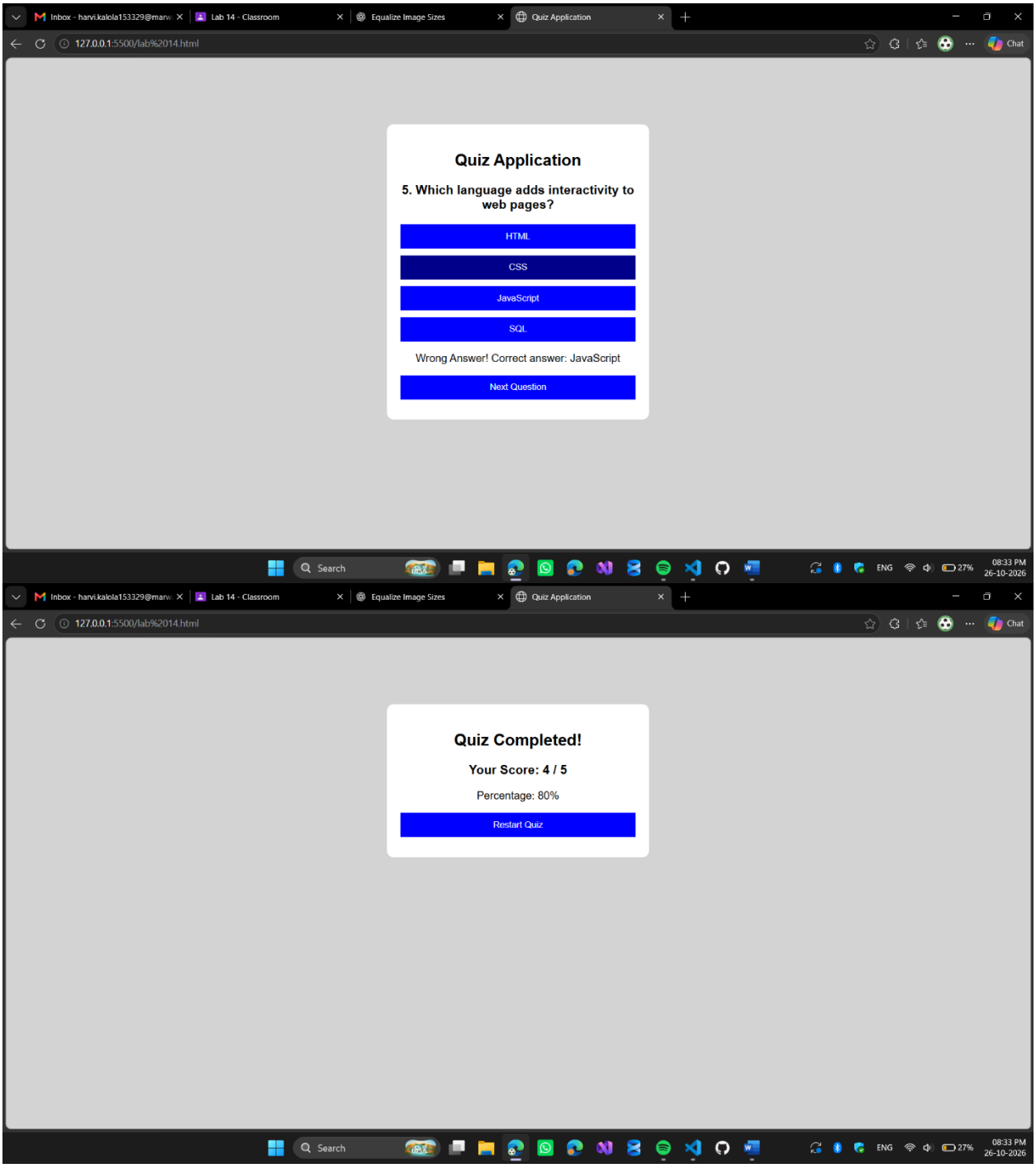
TAGS USED	DESCRIPTION
<!DOCTYPE html>	Declares the document as an HTML5 document.
<html>	Root element of the HTML document.
<head>	Contains metadata and information about the HTML document.
<title>	Specifies the title shown in the browser tab.
<style>	Defines the CSS styles used to design the webpage.
<body>	Contains all visible content of the webpage.
	Groups related elements together in a container.
<h2>	Displays the main heading, "Quiz Application".
<h3>	Displays the current question and final score.
<p>	Displays the answer result and final percentage.
<button>	Creates clickable buttons for answer options, the next question, and restarting the quiz.
<script>	Contains JavaScript code to control the quiz functionality.
const	Declares a constant variable to store the quiz questions and answers.
let	Declares variables whose values can change, such as the current question and score.
function	Defines reusable blocks of JavaScript code.
forEach()	Iterates through the answer options to create buttons.
if...else	Checks whether the selected answer is correct or incorrect.
document.getElementById()	Selects an HTML element using its ID.
document.querySelector()	Selects the first HTML element matching the specified CSS selector.
document.querySelectorAll()	Selects all elements matching the specified CSS selector.
innerHTML	Gets or changes the HTML content of an element.
onclick	Executes a JavaScript function when a button is clicked.
style	Applies inline CSS styling to an HTML element.
disabled	Prevents users from clicking answer buttons after answering a question.
class	Assigns a class name to an HTML element for styling and selection.
id	Assigns a unique identifier to an HTML element.
margin	Sets the space outside an element.

padding	Sets the space inside an element.
background-color	Sets the background color of an element.
color	Sets the text color.
font-family	Specifies the font used for the text.
text-align	Aligns text horizontally.
border-radius	Rounds the corners of an element.
cursor	Specifies the mouse cursor appearance.
:hover	Applies CSS styles when the mouse pointer moves over an element.

Final Outcome:




Quiz Application
3. What is 5 + 5?
8
9
10
11
Correct Answer!
Next Question
Quiz Application
4. Which tag is used for a paragraph in HTML?
<head>
<h1>
<p>
<title>
Correct Answer!
Next Question


Quiz Application
5. Which language adds interactivity to web pages?
HTML
CSS
JavaScript
SQL
Wrong Answer! Correct answer: JavaScript
Next Question
Quiz Completed!
Your Score: 4 / 5
Percentage: 80%
Restart Quiz

Code Snippet:

```
<!DOCTYPE html>
<html>
<head>
  <title>Quiz Application</title>

  <style>
    body {
      font-family: Arial;
      background-color: lightgray;
      text-align: center;
    }

    .quiz {
      background-color: white;
      width: 350px;
      margin: 100px auto;
      padding: 20px;
      border-radius: 10px;
    }

    button {
      display: block;
      width: 100%;
      padding: 10px;
      margin: 10px 0;
      background-color: blue;
      color: white;
      border: none;
      cursor: pointer;
    }

    button:hover {
      background-color: darkblue;
    }
  </style>
</head>
```

```
<body>

  <div class="quiz">
    <h2>Quiz Application</h2>

    <h3 id="question"></h3>

    <div id="options"></div>

    <p id="result"></p>

    <button id="next" onclick="nextQuestion()" style="display:none;">
      Next Question
    </button>
  </div>

  <script>
    const questions = [
      {
        question: "1. What is the capital of India?",
        options: ["Mumbai", "Delhi", "Chennai", "Kolkata"],
        answer: "Delhi"
      },
      {
        question: "2. Which language is used for web pages?",
        options: ["HTML", "C", "Java", "Python"],
        answer: "HTML"
      },
      {
        question: "3. What is 5 + 5?",
        options: ["8", "9", "10", "11"],
        answer: "10"
      },
      {
        question: "4. Which tag is used for a paragraph in HTML?",
        options: ["&lt;head&gt;", "&lt;h1&gt;", "&lt;p&gt;", "&lt;title&gt;"],
        answer: "&lt;p&gt;"
      },
```

```
    {
      question: "5. Which language adds interactivity to web pages?",
      options: ["HTML", "CSS", "JavaScript", "SQL"],
      answer: "JavaScript"
    }
];

let currentQuestion = 0;
let score = 0;

function showQuestion() {
    document.getElementById("result").innerHTML = "";

    let q = questions[currentQuestion];

    document.getElementById("question").innerHTML = q.question;

    let output = "";

    q.options.forEach(function(option) {
        output += `<button onclick="checkAnswer(this)">${option}</button>`;
    });

    document.getElementById("options").innerHTML = output;
}

function checkAnswer(button) {
    let q = questions[currentQuestion];

    if (button.innerHTML === q.answer) {
        score++;
        document.getElementById("result").innerHTML = "Correct Answer!";
    } else {
        document.getElementById("result").innerHTML =
            "Wrong Answer! Correct answer: " + q.answer;
    }

    // Disable all options after answering
    let buttons = document.querySelectorAll("#options button");
```

```
        buttons.forEach(function(btn) {
            btn.disabled = true;
        });

        document.getElementById("next").style.display = "block";
    }

    function nextQuestion() {
        currentQuestion++;

        if (currentQuestion < questions.length) {
            showQuestion();
            document.getElementById("next").style.display = "none";
        } else {
            showResult();
        }
    }

    function showResult() {
        document.querySelector(".quiz").innerHTML =
            "<h2>Quiz Completed!</h2>" +
            "<h3>Your Score: " + score + " / " + questions.length + "</h3>" +
            "<p>Percentage: " +
            Math.round((score / questions.length) * 100) + "%</p>" +
            "<button onclick='location.reload()'>Restart Quiz</button>";
    }

    showQuestion();
  </script>

</body>
</html>
```